Phys 115B HW 7

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1 D

Question 1. *1 Perturbation Theory (Griffiths 5.15, 5.11 in second edition)*

(a) *Calculate* $\left\langle \frac{1}{|\mathbf{r}_1-\mathbf{r}_2|} \right\rangle$ *for the state* $\psi_0 = \frac{8}{\pi a^3} e^{-2(r_1+r_2)/a}$.

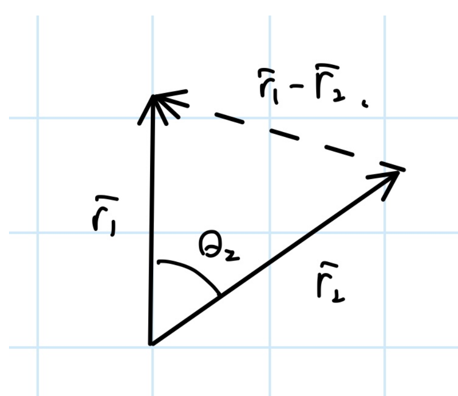
Hint: *Do the* $d^3\mathbf{r}_2$ *integral first, using spherical coordinates, and setting the polar axis along* $\mathbf{r}_1$, *so that* $|\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}$. *The* θ_2 *integral is easy, but be careful to take the* positive root. *You'll have to break the* r_2 *integral into two pieces, one ranging from* 0 *to* r_1, *the other from* r_1 *to* ∞.

Answer: $\frac{5}{4a}$.

(b) *Use your result in (a) to estimate the electron interaction energy in the ground state of helium. Express your answer in electron volts, and add it to* $E_0 = -109$ *eV, to get a corrected estimate of the ground state energy. Compare the experimental value. (Of course, we're still working with an approximate wave function, so don't expect perfect agreement).*

Proof.

(a) Because of the spherical symmetry of ψ_0 (in both $\mathbf{r}_1, \mathbf{r}_2$), then when doing the integration, we can integrate $\mathbf{r}_2$ first, and treat $\mathbf{r}_2$ as in a coordinate where $\mathbf{r}_1$ is in the polar axis:



As a result, using Law of Cosine, one has $\|\mathbf{r}_1 - \mathbf{r}_2\| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}$. So, the full integral can be expanded as follow:

$$\left\langle \frac{1}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right\rangle = \int_{\mathbf{r}_1}\int_{\mathbf{r}_2} \frac{64}{\pi^2 a^6} \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}} e^{-4(r_1+r_2)/a} dV_2 \; dV_1 \tag{1}$$

$$= \frac{64}{\pi^2 a^6} \int_{\mathbf{r}_1} e^{-4r_1/a} \left(\int_{\mathbf{r}_2} \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}} e^{-4r_2/a} dV_2 \right) dV_1 \tag{2}$$

Let's do the integral of $\mathbf{r}_2$ first:

$$I(r_1) := \int_{\mathbf{r}_2} \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}} e^{-4r_2/a} dV_2 \tag{3}$$

$$= \int_{r_2=0}^{\infty} \int_{\phi_2=0}^{2\pi} \int_{\theta_2=0}^{\pi} \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)}} e^{-4r_2/a} r_2^2 \sin(\theta_2) d\theta_2 \; d\phi_2 \; dr_2 \tag{4}$$

$$= 2\pi \int_{r_2=0}^{\infty} \int_{u=(r_1-r_2)^2}^{(r_1+r_2)^2} \frac{1}{\sqrt{u}} e^{-4r_2/a} \frac{r_2}{2r_1} du \; dr_2 \tag{5}$$

$$= \frac{2\pi}{r_1} \int_0^{\infty} r_2 e^{-4r_2/a} \left(\sqrt{u} \Big|_{(r_1-r_2)^2}^{(r_1+r_2)^2} \right) dr_2 \tag{6}$$

$$= \frac{2\pi}{r_1} \int_0^{\infty} r_2 e^{-4r_2/a} ((r_1 + r_2) - |r_1 - r_2|) dr_2 \tag{7}$$

Here, there are two cases:

- First, if $r_1 < r_2$, then $|r_1 - r_2| = r_2 - r_1$, so the sum in the paranthesis is $2r_1$. Which, this results in the following integral:

$$I_< = \frac{2\pi}{r_1} 2r_1 \int_{r_1}^{\infty} r_2 e^{-4r_2/a} dr_2 = 4\pi \left(-\frac{a}{4} r_2 - \frac{a^2}{16} \right) e^{-4r_2/a} \Big|_{r_1}^{\infty} = a\pi \left(r_1 + \frac{a}{4} \right) e^{-4r_1/a} \tag{8}$$

- Then, if $r_1 \geq r_2$, then $|r_1 - r_2| = r_1 - r_2$, so the sum in the paranthesis is $2r_2$ instead. Which, this results in the following integral:

$$I_\geq = \frac{2\pi}{r_1} \int_0^{r_1} 2r_2^2 e^{-4r_2/a} dr_2 = \frac{4\pi}{r_1} \left(-\frac{a}{4} r_2^2 - \frac{a^2}{8} r_2 - \frac{a^3}{32} \right) e^{-4r_2/a} \Big|_0^{r_1} \tag{9}$$

$$= \frac{a^3 \pi}{8r_1} - a\pi \left(r_1 + \frac{a}{2} + \frac{a^2}{8r_1} \right) e^{-4r_1/a} \tag{10}$$

Which, the total $\mathbf{r}_2$ integral is:

$$I(r_1) = I_< + I_\geq = \frac{a^3 \pi}{8r_1} - a\pi \left(\frac{a}{4} + \frac{a^2}{8r_1} \right) e^{-4r_1/a} \tag{11}$$

Then, returning to the original integral, one has the following:

$$\left\langle \frac{1}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right\rangle = \frac{64}{\pi^2 a^6} \int_{\mathbf{r}_1} e^{-4r_1/a} I(r_1) dV_1 \tag{12}$$

$$= \frac{64}{\pi^2 a^6} \int_{r_1=0}^{\infty} \int_{\theta_1=0}^{\pi} \int_{\phi_1=0}^{2\pi} \left(\frac{a^3 \pi}{8r_1} e^{-4r_1/a} - a\pi \left(\frac{a}{4} + \frac{a^2}{8r_1} \right) e^{-8r_1/a} \right) r_1^2 \sin(\theta_1) d\phi_1 \; d\theta_1 \; dr_1 \tag{13}$$

$$= \frac{2^8}{\pi a^6} \int_0^{\infty} \left(\frac{a^3\pi}{8} r_1 e^{-4r_1/a} - \frac{a^2\pi}{4} r_1^2 e^{-8r_1/a} - \frac{a^3\pi}{8} r_1 e^{-8r_1/a} \right) dr_1 \tag{14}$$

$$= \frac{32}{a^3} \int_0^{\infty} \left(r_1 e^{-4r_1/a} - \frac{2}{a} r_1^2 e^{-8r_1/a} - r_1 e^{-8r_1/a} \right) dr_1 \tag{15}$$

$$= \frac{32}{a^3} \left(\left(-\frac{a}{4} r_1 - \frac{a^2}{16} \right) e^{-4r_1/a} - \frac{2}{a} \left(-\frac{a}{8} r_1^2 - \frac{a^2}{32} r_1 - \frac{a^3}{256} \right) e^{-8r_1/a} - \left(-\frac{a}{8} r_1 - \frac{a^2}{64} \right) e^{-8r_1/a} \right) \Bigg|_0^{\infty} \tag{16}$$

$$= \frac{32}{a^3} \left(\frac{a^2}{16} - \frac{a^2}{128} - \frac{a^2}{64} \right) = \frac{1}{a} \left(2 - \frac{1}{4} - \frac{1}{2} \right) = \frac{5}{4a} \tag{17}$$

(b) For Helium, since the Hamiltonian is given as:

$$H = \left(-\frac{\hbar^2}{2m} \nabla_1^2 - \frac{1}{4\pi\epsilon_0} \frac{2e^2}{r_1} \right) + \left(-\frac{\hbar^2}{2m} \nabla_2^2 - \frac{1}{4\pi\epsilon_0} \frac{2e^2}{r_2} \right) + \frac{1}{4\pi\epsilon_0} \frac{e^2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \tag{18}$$

Which, the non-interacting part (the first two paranthesis) has an expectation value of -109 eV, then for the whole expectation value of $\psi_{1,0,0}$, we have:

$$\langle H \rangle = -109 \text{ eV} + \frac{e^2}{4\pi\epsilon_0} \left\langle \frac{1}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right\rangle = -109 \text{ eV} + \frac{5e^2}{16\pi\epsilon_0 a} \tag{19}$$

Which, recall that for Helium ($Z = 2$), one has Bohr radius $\frac{a_0}{2}$ (where a_0 is hte bohr radius of hydrogen), then with electron charge $e = 1.609 \cdot 10^{-19}$ C, and permitivity $\epsilon_0 = 8.854 \cdot 10^{-12}$ $\text{C}^2 \cdot \text{s}^2 / (\text{kg} \cdot \text{m}^{-3})$, and $a = \frac{a_0}{2} = 2.646 \cdot 10^{-11}$ m. As a result, one has the following:

$$\frac{5e^2}{16\pi\epsilon_0 a} = \frac{5 \cdot (1.609)^2 \cdot 10^{-38}}{16\pi \cdot 8.854 \cdot 10^{-12} \cdot 2.646 \cdot 10^{-11}} \text{ J} \approx 1.099 \cdot 10^{-17} \text{ J} \approx 68.317 \text{ eV} \tag{20}$$

So, the actual result is $\langle H \rangle \approx -109 + 68.317 \text{ eV} = -40.683 \text{ eV}$.

In comparison to the experimental value (which is about -79 eV), this is actually higher.

$\square$

2

Question 2. *2 2D Harmonic Oscillator*
Consider the 2D Harmonic Oscillator:

$$V(x,y) = \frac{m\omega x^2}{2} + \frac{m\omega y^2}{2}$$

Define the ground state energy, $E_0 = \hbar\omega_0$. We're going to fill up the 2D harmonic oscillator with particles. Something that might come in handy: the number of ways of distributing N indistinguishable fermions among g sublevels of an energy level with a maximum of 1 particle per sublevel is the binomial coefficient:

$$w(N,g) = \frac{g!}{N!(g-N)!}.$$

1. *What are the energies of the first 6 single particle levels if the particle has spin 0?*
2. *What are the energies of the first 12 single particle levels if the particle has spin 1/2?*
3. *What are the energies of the first 9 single particle levels if the particle has spin 1?*
4. *Now we put in N particles. What is the degeneracy of the ground state with $N = 2$ if the particles are bosons with spin 0?*
5. *What is the degeneracy of the ground state with $N = 2$ if the particles are bosons with spin 1?*
6. *What is the degeneracy of the ground state with $N = 4$ if the particles are fermions with spin 1/2?*
7. *What is the degeneracy of the ground state with $N = 6$ if the particles are fermions with spin 3/2?*

Proof. Here, recall that for a 2D harmonic oscillator, its energy $E_{n,m} = E_{x,n} + E_{y,m} = (n+m+1)\hbar w_0$ for $n, m \in \mathbb{N}$. Which, for any $k \in \mathbb{N}$, if $E_{n,m} = (k+1)\hbar w_0$, one needs $n+m = k$. Hence, for the kth energy level, the number of sublevels in position states g_k = number of ways to partition k into two natural numbers = $k+1$.

1. If there's only a single particle with spin 0 (which only has degeneracy 1 on spin state), then the first 6 levels are just about energies, given by $k = 0$ (with degeneracy $g_0 = 1$), $k = 1$ (with degeneracy $g_1 = 2$), and $k = 2$ (with degeneracy $g_2 = 3$). Which, the corresponding energies are $\hbar w_0, 2\hbar w_0, 3\hbar w_0$ respectively.

2. If the single particle has spin $\frac{1}{2}$ (which the spin state has degeneracy 2), then each energy level will have $2g_k$ sublevels if regarding spins. Hence, the first 12 levels are given by: $k = 0$ (with degeneracy $2g_0 = 2$), $k = 1$ (with degeneracy $2g_1 = 4$), and $k = 2$ (with degeneracy $2g_2 = 6$). So, the corresponding energies are $\hbar w_0, 2\hbar w_0, 3\hbar w_0$ respectively.

3. If the single particle has spin 1 (which the spin state has degeneracy 3), then each energy level will have $3g_k$ sublevels if regarding spins. Hence, the first 9 levels are given by: $k = 0$ (with degeneracy $3g_0 = 3$), and $k = 1$ (with degeneracy $3g_1 = 6$). So, the corresponding energies are $\hbar w_0$, and $2\hbar w_0$ respectively.

4. Given that the bosons are spin 0 (spin state has degeneracy 0), with $N = 2$, since they can all be packed in the same energy level, so the ground state of the system has all of them with ground state $k = 0$. Which, the degeneracy is in fact 1 (since ther's no choice on the energies or the states).

5. Given the bosons are spin 1 (spin state has degeneracy 3), with $N = 2$, since they can all be packed in the same energy level, so the ground state of the system has all of them with ground state $k = 0$, hence the position side there's degeneracy 1 (in particular, it's symmetric in position states). Now, this enforces the spin states to be symmetric also. If regarding the spin side, since for two 3-dimensional Hilbert Space, the symmetric tensor has dimension $\frac{4 \cdot 3}{2} = 6$, so spin state has degeneracy 6. Hence, the total degeneracy is 6.

6. Given the particles are frmions with spin $\frac{1}{2}$ (spin state has degeneracy 2), since the top wedge of a 2-dimensional Hilbert Space can consist of only 2 elements (i.e. one can only have wedge-2 product, for wedge $n \geq 3$ the wedge product is 0), each energy sublevel is only allowed with 2 such fermions (since with position states being symmetric, the spin states are antisymmetric, which must be in the wedge product; the top wedge represents the number of particles one can put in the same energy sublevel and still maintain the antisymmetric property).

   With $N = 4$, the first two fermions can be put into the ground state $k = 0$, and with the symmetric position states, their spin state is antisymmetric (for spin-$\frac{1}{2}$), which has degeneracy 1.

   For the last two fermions, since there are $g_1 = 2$ sublevels for energy level $k = 1$, then it's posible to put all of them in energy level $k = 1$. There are two cases: If both fermions are in the same sublevel, then the symmetric position state enforces antisymmetric spin states, which has degeneracy 1 (and since there are 2 distinct sublevels, there are 2 such cases); else, if the fermions are belonging two distinct sublevels, then each occupies a spin state of degeneracy 2, hence has a total degeneracy of $2 \cdot 2 = 4$. Then, the two fermions in energy level $k = 1$ in fact has degeneracy $2 + 4 = 6$.

   Combining all information, this system has ground state degeneracy of 6.

7. If the particles are fermions with spin $\frac{3}{2}$ (spin state has degeneracy 4), since the top wedge of a 4-dimensional Hilbert Space can consist of 4 elements (i.e. one can have wedge-4 product, and for wedge $n \geq 5$ the wedge product is 0), each energy sublevel is allowed with 4 such fermions.

   With $N = 6$, the first four fermions can be put into the ground state $k = 0$, and with the symmetric position states, the spin state is antisymmetric (with 4 particles and spin-$\frac{3}{2}$), which again has degeneracy 1.

   For the last two fermions, since there are $g_1 = 2$ sublevels for energy level $k = 1$, it's possible to put the in $k = 1$ energy level. There are two cases: If both fermions are in the same sublevel, the symmetric position state enforces antisymmetric spin states, with the spin state described by two

4-dimensional Hilbert Space, the wedge product has dimension $\frac{4 \cdot 3}{2} = 6$, hence it has degeneracy 6 (and with two distinct sublevels, the total degeneracy is 12); else, if the fermions are belonging to two distinct sublevels, then each occupies a spin state of degeneracy 4, hence has a total degeneracy of $4 \cdot 4 = 16$. Then, the two fermions in energy level $k = 1$ in fact has degeneracy $12 + 16 = 28$.

Combining all information, this system has ground state degeneracy of 28.

$\square$

3 D

Question 3. *3 Silver (modified Griffiths 5.21, 5.16 in second edition)*
The density of silver is 10.49 g/cm^3, and its atomic weight is 107.9 g/mole.

(a) *Calculate the Fermi energy for silver. Assume* $d = 1$ *and give your answer in electron volts.*

(b) *What is the corresponding electron velocity? Hint: Set* $E_F = \frac{1}{2}mv^2$*. Is it safe to assume the electrons in silver are nonrelativistic?*

(c) *At what temperature* T_F *would the characteristic thermal energy (*k_BT*, where* k_B *is the Boltzmann constant and* T *is the temperature in Kelvin) equal the Fermi energy for silver? This is called the Fermi temperature* T_F*. As long as the actual temperature is significantly below* T_F*, the material can be regarded as "cold" in the sense that most of the electrons are in the lowest accessible state. Since the melting point of silver is 1235 K, solid silver is always cold.*

(d) *Calculate the degeneracy pressure of silver, in the free electron gas model.*

Proof.

(a) Recall the Fermi Energy is $E_F := \frac{\hbar^2}{2m}k_F^2 = \frac{\hbar^2}{2m}(3\rho\pi^2)^{\frac{2}{3}}$, and ρ represents the number of electrons per unit volume (say in terms of m^3). After conversion of units, one has silver's density $10.49 \cdot 10^6$ g/m^3, hence the silver density can also be written as $10.49 \cdot 10^6/107.9$ mole/m^3.

With each mole having $6.02 \cdot 10^{23}$ atoms, there are total of $10.49 \cdot 10^6 \cdot 6.02 \cdot 10^{23}/107.9$ number of silver atoms per m^3, which is about $5.853 \cdot 10^{28}$ number of silver atoms per m^3; with silver having only one valence electron, this is also the number of free electrons in silver per m^3, or ρ.

Now, given that $\hbar = 1.055 \cdot 10^{-34}$ kg·m^2/s, and the mass of an electron $m_e = 9.109 \cdot 10^{-31}$ kg. Pllugin to the Fermi Energy equation, we have:

$$E_F = \frac{(1.055 \cdot 10^{-34})^2}{2 \cdot 9.109 \cdot 10^{-31}}(3\pi^2)^{\frac{2}{3}}(5.853 \cdot 10^{28})^{\frac{2}{3}} \approx 8.816 \cdot 10^{-19}J \approx 5.509eV \tag{21}$$

(b) Suppose $E_F = \frac{1}{2}mv^2$, with $m_e = 9.109 \cdot 10^{-31}$ kg, then the velocity becomes:

$$v = \sqrt{\frac{2E_F}{m_e}} = \sqrt{\frac{2 \cdot 8.816 \cdot 10^{-19}}{9.109 \cdot 10^{-31}}} \approx 1.936 \cdot 10^6 \text{ m/s} \tag{22}$$

Which, this is approximately 0.6% of light speed, it can possibly considered as relativistic.

(c) Given the Boltzmann constant $k_B = 1.381 \cdot 10^{-23}$ J/K, if temperature T_F satisfies $E_F = k_BT_F$, then:

$$T_F = \frac{E_F}{k_B} \approx \frac{8.816 \cdot 10^{-19}}{1.381 \cdot 10^{-23}} = 6.384 \cdot 10^4 \text{ K} \tag{23}$$

(d) Given the degeneracy pressure $P = \frac{2}{3}\frac{E_{\text{tot}}}{V} = \frac{(3\pi^2)^{2/3}\hbar^2}{5m}\rho^{5/3}$. Which, the detail of the formula provides:

$$P = \frac{(3\pi^2)^{2/3}(1.055 \cdot 10^{-34})^2}{5 \cdot 9.109 \cdot 10^{-19}}(5.853 \cdot 10^{28})^{5/3} \approx 0.0206 \text{ Pa} \tag{24}$$

□

4 D

Question 4. *4 2D Gas Model (Griffiths 5.30, 5.34 in the second edition)*
Calculate the Fermi energy of the electrons in a two-dimensional infinite square well. Let σ be the number of free electrons per unit area.

Proof. Given a two-dimensional infinite square well where $x \in [0, L_x]$ and $y \in [0, L_y]$, then a single electron has allowed eigenstates being $\psi_{n_x,n_y}(x,y) = \sqrt{\frac{4}{L_x L_y}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right)$, which has energy $E = E_x + E_y = \frac{\hbar^2 \pi^2}{2m}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2}\right)$. Then, given wavevector $\mathbf{k} = \left(\frac{\pi n_x}{L_x}, \frac{\pi n_y}{L_y}\right)$, one has energy $E = \frac{\hbar^2}{2m}k^2$, where $\|\mathbf{k}\| = k$.

Since every increment in n_x or n_y represents an allowed eigenstate, which $k_x = \frac{n_x \pi}{L_x}$ and $k_y = \frac{n_y \pi}{L_y}$ incremented by $\frac{\pi}{L_x}$ and $\frac{\pi}{L_y}$ respectively. Then, each eigenstate "occupies" a volume of $\frac{\pi}{L_x} \cdot \frac{\pi}{L_y} = \frac{\pi^2}{L_x L_y}$ inside the $\mathbf{k}$-space. With the area of the well being $A := L_x L_y$, each volume represents $\frac{\pi^2}{A}$.

Now, for the minimized energy of electron gases in an infinite square well, in the $\mathbf{k}$-space it reduces to a quarter circle (as the energy must have $k_x, k_y \geq 0$), hence for the number of occupied states (say N), one has $N \cdot \frac{\pi^2}{A} \approx \frac{1}{4}\pi k_F^2$ (where k_F is the radius of the sphere that has the occupied states). Hence, $N \approx \frac{A}{4\pi^2}k_F^2$.
Then, given many electrons, with the Pauli Exclusion Principle guaranteeing each eigenstate to have at most 2 electrons, then the number of electrons, say I, satisfies $I \approx 2N \approx \frac{A}{2\pi^2}k_F^2$. Hence, one derived that $k_F^2 \approx 4\pi^2 \frac{N}{A} = 4\pi^2 \sigma$ (since $\frac{N}{A}$ is the number of free electrons per unit area), so $k_F \approx 2\pi\sqrt{\sigma}$.

As a result, the Fermi Energy $E_F := \frac{\hbar^2}{2m}k_F^2 = \frac{2\pi^2 \hbar^2 \sigma}{m}$. $\square$

5 D

Question 5. *5 White Dwarfs (Griffiths 5.35 in both)*
Certain cold stars (called ***white dwarfs****) are stabilized against gravitational collapse by the degeneracy pressure of their electrons ($P = \frac{2}{3}\frac{E_{tot}}{V} = \frac{2}{3}\frac{\hbar^2 k_F^5}{10\pi^2 m} = \frac{(3\pi^2)^{2/3}\hbar^2}{5m}\rho^{5/3}$). Assuming constant density, the radius R of such an object can be calculated as follow:*

(a) Write the total electron energy ($E_{tot} = \frac{\hbar^2 V}{2\pi^2 m}\int_0^{k_F} k^4 dk = \frac{\hbar^2 k_F^5 V}{10\pi^2 m} = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{10\pi^2 m}V^{-2/3}$) in terms of the radius, the number of nucleons (protons and neutrons) N, the number of electrons per nucleon d, and the mass of the electron m.

Beware: *In this problem we are recycling the letters N and d for slightly different purpose than in the text.*

(b) Look up, or calculate, the gravitational energy of a uniformly dense sphere. Express your answer in terms of G (the constant of universal gravitation), R, N, and M (he mass of a nucleon). Note that the gravitational energy is negative.

(c) Find the radius for which the total energy, (a) plus (b), is minimum.

Answer: $R = \left(\frac{9\pi}{4}\right)^{2/3}\frac{\hbar^2 d^{5/3}}{GmM^2N^{1/3}}$.

(Note that the radius decreases *as the total mass* increasees*!).*

Put in the actual numbers, for everything except N, using $d = \frac{1}{2}$ (actually, d decreases a bit as the atomic number increases, but this is close enought for our purposes).

Answer: $R = 7.6 \times 10^{25} N^{-1/3}$ *m.*

(d) Determine the radius, in kilometers, of a white dwarf with the mass of the sun.

(e) Determine the Fermi energy, in electron volts, for the white dwarf in (d), and compare it with the rest energy of an electron. Note that this system is getting dangerously relativistic.

Proof.

(a) Given the radius of the white dwarf R, then the volume of a ball is $\frac{4}{3}\pi R^3$. Hence, $V^{-2/3} = \left(\frac{3}{4\pi}\right)^{2/3}\frac{1}{R^2}$. Therefore, the total energy is of this form:

$$E_{\text{tot}} = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{10\pi^2 m}V^{-2/3} = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{10\pi^2 mR^2}\left(\frac{3}{4\pi}\right)^{2/3} \tag{25}$$

(b) Fir a constant density ρ for the ball of radius R. Given a spherical shell with radius r and thickness Δr (which has volume $\approx 4\pi r^2\Delta r$). Then with respect to the ball of constant mass in the middle (with volume $\frac{4}{3}\pi r^3$), the gravitational potential energy is given as follow:

$$\Delta U = \frac{Gm_1m_2}{r} = \frac{G}{r}\left(\rho\cdot 4\pi r^2\Delta r\right)\cdot\left(\rho\cdot\frac{4}{3}\pi r^3\right) = \frac{16\pi^2 G\rho^2}{3}r^4\Delta r \tag{26}$$

Hence, the whole ball would obtain the following gravitational energy in terms of density:

$$U = \int_0^R \frac{16\pi^2 G\rho^2}{3}r^4 dr = \frac{16\pi^2 G\rho^2}{15}R^5 \tag{27}$$

Now, given the total volume $V = \frac{4}{3}\pi R^3$, if there are N nucleons, with each nucleon taking mass M, then the density $\rho = \frac{NM}{V} = \frac{3NM}{4\pi R^3}$. Plugin to the equation, we get:

$$U = \frac{16\pi^2 G R^5}{15} \cdot \left(\frac{3NM}{4\pi R^3}\right)^2 = \frac{3GN^2M^2}{5R} \tag{28}$$

If consider gravity as negative potential, then $U = -\frac{3GN^2M^2}{5R}$ instead.

(c) If consider the total energy, it's going to be of the form:

$$E_{\text{tot}} + U = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{10\pi^2 m}\left(\frac{3}{4\pi}\right)^{2/3}\frac{1}{R^2} - \frac{3GN^2M^2}{5}\frac{1}{R} \tag{29}$$

For the minimum (given that $R > 0$), it'll occur when the derivative is 0:

$$\frac{d}{dR}(E_{\text{tot}} + U) = -\frac{\hbar^2(3\pi^2 Nd)^{5/3}}{5\pi^2 m}\left(\frac{3}{4\pi}\right)^{2/3}\frac{1}{R^3} + \frac{3GN^2M^2}{5}\frac{1}{R^2} = 0 \tag{30}$$

$$\implies \frac{3GN^2M^2}{5}R = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{5\pi^2 m}\left(\frac{3}{4\pi}\right)^{2/3} \tag{31}$$

$$\implies R = \frac{\hbar^2(3\pi^2 Nd)^{5/3}}{3\pi^2 mGN^2M^2}\left(\frac{3}{4\pi}\right)^{2/3} = \left(\frac{9\pi}{4}\right)^{2/3}\frac{\hbar^2 d^{5/3}}{mGM^2N^{1/3}} \tag{32}$$

Which, given $d = \frac{1}{2}$, nucleon mass $M = 1.67 \cdot 10^{-27}$ kg, reduced Planck's constant $\hbar = 1.055 \cdot 10^{-34}$ kg·m^2/s, electron mass $m = 9.109 \cdot 10^{-31}$ kg, and universal gravitational constant $G = 6.674 \cdot 10^{-11}$ m^3/(kg·s^2), then the radius becomes:

$$R = \left(\frac{9\pi}{4}\right)^{2/3}\frac{(1.055)^2 \cdot 10^{-68} \cdot 2^{-5/3}}{9.109 \cdot 6.674 \cdot (1.67)^2 \cdot 10^{-96}}\frac{1}{N^{1/3}} \approx \frac{7.616 \cdot 10^{25}}{N^{1/3}} \text{ m} \tag{33}$$

(d) Given total number of N nucleons, with $d = \frac{1}{2}$ as assumption, then the total mass of the system is given as:

$$M_{\text{tot}} = N\left(M + \frac{1}{2}m\right) = N\left(1.67 \cdot 10^{-27} + 4.555 \cdot 10^{-31}\right) \text{ kg} \approx 1.67045 \cdot 10^{-27} \text{ kg} \tag{34}$$

If it's the mass of the sun, $M_{\text{tot}} = 1.989 \cdot 10^{30}$ kg, then the number N should be given as:

$$N = \frac{M_{\text{tot}}}{1.67045 \cdot 10^{-27}} \approx 1.191 \cdot 10^{57} \tag{35}$$

Plugin to the radius formula, one yields:

$$R \approx \frac{7.616 \cdot 10^{25}}{(1.191 \cdot 10^{57})^{1/3}} \text{ m} \approx 7184.938 \text{ km} \tag{36}$$

(e) Recall that the Fermi energy $E_F = \frac{\hbar^2}{2m}(3\rho\pi^2)^{2/3} = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 Nd}{V}\right)^{2/3}$. Given $d = \frac{1}{2}$, $N \approx 1.191 \cdot 10^{57}$, $\hbar = 1.055 \cdot 10^{-34}$ J·s, $m = 9.109 \cdot 10^{-31}$ kg, and $R \approx 7.184938 \cdot 10^6$ m. Then, the volume $V = \frac{4}{3}\pi R^3 \approx 1.554 \cdot 10^{21}$ m^3, and the resulting Fermi energy is:

$$E_F = \frac{(1.055)^2 \cdot 10^{-68}}{2 \cdot 9.109 \cdot 10^{-31}}\left(\frac{3\pi^2 \cdot 1/2 \cdot 1.191 \cdot 10^{57}}{1.554 \cdot 10^{21}}\right)^{2/3} \approx 3.085 \cdot 10^{-14} \text{ J} \approx 1.928 \cdot 10^5 \text{ eV} \tag{37}$$

If comparing to the rest energy of an electron, one gets:

$$E_{\text{rest}} = mc^2 = (9.109 \cdot 10^{-31}) \cdot 9 \cdot 10^16 \approx 8.198 \cdot 10^{-14} \text{ J} \approx 5.124 \cdot 10^5 \text{ eV} \tag{38}$$

So, the Fermi energy is slightly less than the rest mass energy, but they're about the same scale. □